QP Code **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-BVOC (HONOURS) Regular EXAMINATION, October 2026

Third SEMESTER

Skill Development Components (SDC) – MG3SDCFTA202 – FOOD CHEMISTRY

(2025 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

One or two sentences

Answer any **10** questions

Each question carries **1** marks

1. Define free water. [K] / [CO1]
2. Give a note on water activity. [K] / [CO1]
3. Draw the open chain structure of fructose. [K] / [CO1]
4. Define epimers. [K] / [CO1]
5. Give a note on gelation. [K] / [CO1]
6. List any two aromatic amino acid with its structure. [K] / [CO2]
7. Give a note on peptide bond. [K] / [CO2]
8. Describe zwitter ion. [K] / [CO2]
9. Distinguish fat and oil. [K] / [CO3]

10. Define the term polymorphism. [K] / [CO3]

11. List out the factors that affect the plasticity of fat. [K] / [CO3]

12. Define RM value. [K] / [CO3]

[1x10 = 10]

Part B

Short answer

Answer any **4** questions

Each question carries **5** marks

13. Discuss on starch gelatinization [K] / [CO1]

14. Discuss on enzymatic browning and its method of control [K] / [CO1]

15. Classify protein based on source [K] / [CO2]

16. Discuss on protein denaturation. [K] / [CO2]

17. Recognize the difference between smoke, flash and fire point. [K] / [CO3]

18. Explain hydrogenation of oils. [U] / [CO3]

[5x4 = 20]

Part C

Essay

Answer any **2** questions

Each question carries **10** marks

19. Elaborate the structural classification of carbohydrates with examples. [K] / [CO1]

20. Discuss on Maillard reaction and Caramelization. [U] / [CO1]

21. Explain in detail on protein structure. [U] / [CO2]

22. Discuss the classification of edible fat. [U] / [CO3]

[10x2 = 20]